

AI Driven Smart Health Advisor

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Abstract—A Smart Health Advisor based on AI is suggested to handle the two-fold problem of managing chronic diseases and responding to epidemics in isolated, resource-limited populations by redirecting the emphasis away from the reactive process of care, namely sick care, to preventive care. The system uses the Internet of Medical Things (IoMT) and Artificial Intelligence (AI) to constantly analyze real-time information provided by wearable sensors and past medical data, giving personalized 24 / 7 health advice. The platform is supported by a two-engine Python architecture; a predictive engine processes user vitals like blood pressure and glucose levels with the help of mathematical logic that classifies chronic health risk and a regulatory inquiry engine is an engine that maps user query to proscribed protocols in bodies like WHO and NCDC with the help of Natural Language Processing. The design will make users directly connected to current evidence-based precautions and safety measures, which will in turn minimize the proliferation of medical misinformation in cases of emergency. The system is deliberately lightweight and modular in order to be able to run on the bare minimum of hardware and be usable in low-connectivity area with limited connectivity typical of underserved areas. As the experimental analysis suggests, the advisor is consistent in drawing attention to a high-risk situation, such as a sharp spike in blood pressure and clusters of symptoms related to a viral outbreak such as Nipah or COVID-19, and sends direct notifications to the user. A combination of vital analysis with the authoritative knowledge of the regulation results in a structured decision tree that helps the user to understand when the situation can be effectively handled at home or when an immediate professional intervention is needed. This combined methodology unites automated health diagnostics with formal medical communication, helps in the long-term chronic care and immediate emergency response on a single platform. Consequently, the AI-based Smart Health Advisor can boost health literacy, better resource distribution, and empower the patients, especially in the underserved and isolated communities, to make informed decisions, which can eventually result in life-saving.

Keywords—Artificial Intelligence; Preventive Care; Smart Health Advisor; Internet of Medical Things (IoMT); Chronic Disease Risk Prediction; Regulatory Health Informatics

I. INTRODUCTION

The past ten years have seen an unprecedented increase in the prevalence rates of chronic non communicable disease and outbreaks of different infectious diseases among the health care systems across the globe. As well as the ongoing monitoring and self-management of hypertension and diabetes, the emergent threat of Nipah and COVID 19 are causing the need to immediately seek out verified and updated clinical guidelines. The latter burden is particularly compound when it comes to remote or underserved areas, where physician access, as well as diagnostic centers and dependable health information are scarce. The patients in these

environments should have to make vital decisions regarding their health relying on the half-complete, old or un-proven information.

The conventional patterns of delivering healthcare are highly reactive and episodic whereby people usually engage with the health system once the symptoms have gotten critical. This method postpones the timely risk identification, doubles unnecessary hospitalization, and burden already normal medical facilities. The situation becomes even worse in case of an outbreak: on-the-job personnel have to cope with a stream of new cases, health-related messages are rapidly adjusted, and fake news disseminates across social media and informal connections. Consequently, they leave most individuals neglecting very crucial precautions or end up clogging hospitals with cases that would have been handled properly at home. Meanwhile, IoMT, low cost sensors, and Artificial Intelligence (AI) have made it possible to continuously record and process physiological indicators and situational health data outside traditional clinical environments, but it has not been expanded to a single platform of preventive care.

The main driving force behind this is the need to transform healthcare into proactive preventive care and not intervention at a late stage without the use of advanced infrastructure. When people have the chance to have the advisory information on their personal devices on time, clear and trusted, they can prevent most of the negative occurrences that accompany chronic diseases before the situation worsens to an emergency. As an illustration, a consistent high blood pressure reading might lead to systematic alerts and a lifestyle change way before it is too late and the organ may be damaged or the risk of stroke develops. The second reason is the necessity to fight the misinformation regarding health, which is especially perilous at the time of epidemic as humans rely on unverified social information to make their decisions. A clear requirement is the need of a digital intermediary that links the end users directly with protocols that are issued by the authoritative organs like the World Health Organization (WHO) and national disease control institutions that converts technical instructions into context sensitive and comprehensible recommendations.

Current AI applications in health currently exist primarily as symptom checkers, tele-consultation applications or disease-specific monitoring applications, all focusing only on a small section of the problem. Symptom checkers seldom combine ongoing vitals or reconcile their suggestions with current regulatory guidance, teleconsultation continues to rely on limited clinical staff and disease specific applications rarely take into account epidemic co-morbidities or generalized triage protocols. A large number of automated systems compute risk using vitals or symptoms but fail to model WHO

or national ministry-specific guidelines into their reasoning, and thus offer generic or obsolete advice. Research prototypes often have an assumption of stable broadband connection and high end smart phones and this has left an unsolved gap on solutions that are optimized to low resource environments, with lightweight, interpretable logic and minimal computing but still provide integrated decision support.

To bridge these gaps, this paper proposes an AI driven Smart Health Advisor using a dual engine architecture which has been developed using Python. The former engine is aimed at preventive chronic care, taking the user reported vitals (blood pressure and glucose) and classifying the risks based on deterministic mathematical rules that maintain low hardware requirements and interpretable output. The second engine does regulatory health informatics, which is basically the use of Natural Language Processing to the questions of the users and match them to the latest guidelines provided by WHO and the country health ministries, thereby decreasing reliance on unverified information sources. The overall software design is made lightweight, modular and robust to intermittent connectivity to allow it to run on simple hardware regularly available in underserved areas. The proposed Smart Health Advisor will help to reduce the typical misinformation, emphasize the situations when life or health at risk, and state when it is time to seek professional medical attention and when home care is adequate by providing predictive vital analysis and authoritative regulatory knowledge into one advisory structure, thereby allowing to transition the populations that require it to the practice of sick care to preventive care.

II. LITERATURE REVIEW

The rationale for the application of AI in a clinical setting has been well established. Machine learning and rule-based systems can aid in the accuracy of diagnoses and triage if the system has access to the necessary information. This has been known for several years. What has been built upon has been the next logical step in the progression and evolution of the technology. If the technology has the capability to be placed on the patient and transmit information in real time, then the technology has the capability to identify problems before they become a crisis. Wearables and IoMT technologies allow the system to be accomplished with heart rate, glucose levels, and blood pressure. The validation on the technology has been consistent in a clinical setting. Outside of a clinical setting, the infrastructure necessary to accomplish the application of the technology makes it difficult to deploy the system in a setting where adequate healthcare infrastructure does not already exist. This is somewhat ironic when considering the application of the technology in serving the underserved populations.

The application and development of symptom checkers took a different path. One might say it was a more practical application. They didn't require the infrastructure. They simply require the user to input the symptoms, and the system will map the possible conditions. Then the user will be able to determine if they should wait or see a doctor. Some of the tools even go further and allow the user to predict diseases in order to self-diagnose and avoid illness. Some tools even allow the user to receive suggestions on the best course of treatment directly from the system. This has been user-

centric. They are fast and easy to use. They are only as accurate as the user. The logic in the system has been somewhat static. The information the system gives to the user was built in when the system was initially launched and has never been altered.

The NLP-based chatbots have similar terrain but traverse it in a conversational manner. Better ones include medication information and lifestyle suggestions with the help of medical NLP. Things took a whole different turn for the worse with the advent of large language models, which can give confident and articulate health advice even if it is wrong. A study published in 2026 attempted to chart the manner in which LLMs spread medical misinformation in a systematic manner. Needless to say, the results were for the worse. Related studies done with the help of retrieval augmented generation and context protocols for answering clinical questions are beginning to chart the manner in which they perform well and in which they perform badly. One thing that none of these tools, old or new, does is access information from the WHO or country-specific disease control organizations. This lag between the officially recommended information and the information being passed on by the chatbot makes a big difference in a pandemic situation. Remote monitoring is the technically advanced end of this spectrum. They can analyze constant flows of information in order to detect slowly evolving trends, like blood pressure increasing gradually or physical activity levels decreasing gradually. Some have shown promise in sending real-time alerts to clinical staff with the help of 5G-based IoMT networks. They perform well in a clinical setting. A mixed-methods study published in 2026 revealed that performance was affected considerably in the real-world low-resource setting, which complicates the assumption that these tools will be beneficial in regions where health care is scarce.

There are a few other things that intersect with all of the above. With regard to interpretability, there is comparative research between neural networks and decision trees in low-resource settings that suggests the latter still have real advantages in terms of transparency and auditability, even if they slightly underperform in terms of prediction [9], [18]. With regard to AI triage more broadly, systematic reviews confirm the potential of AI tools in speeding up triage decisions [16], although the equity and safety work in real-world deployment settings clearly highlights the potential of AI tools in actually reinforcing existing disparities if not well managed [13]. A scoping review of AI agents in healthcare up to 2026 found the proliferation of agentic architectures in healthcare with governance structures that have not kept pace [15]—something that the WHO has directly addressed in terms of technical guidelines around the deployment of multi-agent AI in public health settings [19]. Forward-looking research suggests the continued development of decentralized AI and blockchain-based health data management up to 2026 [10], although the extent to which these advances will be equitably beneficial remains tied to the unresolved infrastructure gap.

If we take a step back from this list, we can see that there was consistency in the gaps. The symptom checkers were present

but only provided qualitative information. The IoMT tools provided access to rich numerical information but required institutional infrastructure. No tool was present in the existing landscape that provided both chronic disease management and infectious disease surveillance within the same system. The advice was provided based on statistical patterns rather than specific and version-controlled guidelines. This was okay since there were no evolving guidelines. The decision logic was opaque in several systems such that neither patients nor doctors could audit what advice was provided, nor reasons for such advice [14], [15].

This system attempts to fill all of these gaps. The system utilizes a predictive engine based on Python to evaluate chronic disease risk based on objective vitals such as blood pressure and glucose levels. The system utilizes transparent mathematical logic rather than a black box. The system utilizes a regulatory query engine based on Node.js to directly map user queries to verified WHO and NCDC protocols. The system operates on low-resource hardware without cloud infrastructure and 5G. The system provides both chronic care and epidemic response within the same system. The question is, does this system actually meet all of these criteria? The remainder of this paper answers this. In summary, as can be seen from Table 1 below, the Proposed Advisor adopts WHO guidelines as compared to standard LLMs.

System Feature Matrix

FEATURE	TRADITIONAL CHECKERS	STANDARD LLMs	PROPOSED ADVISOR
Triage Logic	Self-reported symptoms only	Probabilistic / General training	Combined Vitals (BP/BMI) & Symptoms
Regulatory Alignment	Internal company rules	No verifiable sourcing	100% WHO/NCDC Guideline RAG Integration
Epidemic Awareness	Rarely updated	Cut-off knowledge dates	Live WHO RSS Disease Outbreak Feeds
Emergency Routing	Generic "Go to hospital"	Generic "Seek medical care"	Dynamic Nomination/OSM Google Maps Routing
Architecture	Heavy monolithic apps	Cloud-heavy, high latency	Lightweight FastAPI Microservices

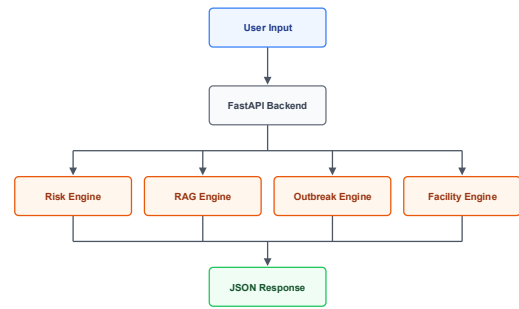
III. METHODOLOGY

The designed AI based Smart Health Advisor will utilize a dual engine, modular design; incorporating chronic risk anticipation with outbreak conscious regulatory advice, with the implementation of lightweight web and AI technologies. This section provides an overview of the design as well as details the predictive and regulatory elements and the manner in which the selected methods advance on current low bandwidth, low resource environments approaches.

A. System architecture

The system has three layers; user interface, orchestration and two inference engines and is based on service oriented approach. It is a react based web interface that gathers numerical vitals and natural language descriptions of symptoms using simple forms allowing it to use on low end devices and limited networks. The Express.js server is used as the key router, directing the numerical data to the Python based Predictive Health Engine and free text-based queries to the Regulatory Inquiry Engine. Here is an illustration of the transition from the user input using React technology through the FastAPI and Node.js inference engines. Here is an illustration of the transition from the user input using React

technology through the FastAPI and Node.js inference engines.



The Predictive Health Engine is deployed as a Python FastAPI microservice, which provides a lightweight REST API to run on a small hardware. Scikit learn offers explainable models in the form of Decision Trees and Random Forests, although the first prototype is based mainly on clear mathematical thresholds. The Regulatory Inquiry Engine is based on the Node.js platform and LangChain.js to coordinate a Retrieval Augmented Generation (RAG) pipeline on top of internal WHO and NCDC health protocols in MongoDB Atlas Vector Search. The two engines have open endpoints that Express.js can invoke, and scale-independent and deploy flexibility. The suggested architectural design adopts microservices as opposed to heavy monolithic applications (see Table 1).

B. Predictive Health Engine

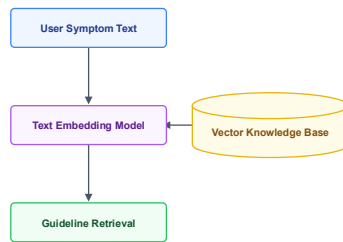
The Predictive Health Engine is aimed at preventive care in the long term by processing the blood pressure and fasting glucose data entered by the user. It uses deterministic clinical thresholds using standard hypertension and diabetes guidelines to categorize readings into the normal, pre risk and high risk band without undergoing training of complex machine learning models. Once the values move above safety margins, the engine sets a high risk flag and develops tangible recommendations including lifestyle change and professional assessment schedules.

The engine provides a plug-in layer, which is powered by scikit learn, so that Decision Trees or Random Forests can be trained on anonymized data and explainability can be maintained through the extraction of rules and feature importance. This approach gives more focus to device or low resource inference, simple thresholds on critical alerts and optional ML extensions to richer analytics compared to black box neural models typically implemented on cloud platforms. This is a direct response to the high compute resource needs of most IoMT based predictive systems and makes the engine more ideal in rural and bandwidth limited environments.

C. Regulatory Inquiry Engine

Regulatory Inquiry Engine deals with outbreak aware triage and control of misinformation. Express.js passes the texts to the Node.js NLP pipeline when users enter natural language descriptions of their symptoms or concerns. First step processing is based on either a tokenizing or a key word or pattern match to quickly identify high priority conditions such as severe fever, respiratory distress or named zoonotic diseases like Nipah.

In more complicated queries, LangChain.js organizes a RAG workflow: it integrates the query and queries MongoDB Atlas Vector Search, which is a database of vectorized WHO and NCDC protocols. Passages retrieved are condensed into brief answers that maintain essential requirements, including isolation, contact tracing, or environmental precautions (e.g. no contact with fruit bats or boiling date palm sap in Nipah areas). This is unlike most health chatbots that make use of a set of rules or generic medical information without a clear connection to existing regulatory documents. The encoding of some outbreak instructions, including a 21 day tracking period on close contacts and immediate isolation on high risk clusters of symptoms, has been codified as system logic. This graph shows the process involved in the transition of free-form text queries into evidence-based recommendations



D. Comparison, restrictions and novelty.

As compared to the traditional symptom checkers that aligns the answer on the questionnaires with the probability of a disease, the developed methodology will integrate structured vitals analysis with active retrieval of regulatory texts balancing risk assessment of the physiological with policy congruent advice. In contrast to cloud heavy IoMT systems, it focuses on a Python FastAPI microservice and threshold based logic and optional lightweight models, which lowers the computational burden and simplifies its deployment to simple infrastructure. The LangChain based RAG over vectorized WHO and NCDC text bases the recommendations on the existing public health guidelines, making it more reliable than a rule based chatbot.

The existing shortcomings are the reliance on correct manual entry of vital signs that can result in measurement inaccuracies with respect to automated integration in IoMT. The threshold focused baseline might be under fitting multi factor interactions that are potentially present in richer ML models but the architecture permits such models to be added progressively. To have an up to date protocol corpus, it is necessary to consume new WHO and NCDC documents regularly and the quality of RAG hinges on their coverage and clarity.

The key methodological innovation is that it offers a single unified advisory process that combines a low resource, explainable predictive engine with a regulation grounded NLP engine. The system provides the output of chronic risk assessment and outbreak specific advice without requiring large-end hardware or round-the-clock supervision of clinicians by passing vitals and languages through coordinated microservices. The close integration of

LangChain coordinated RAG with real time triage, and the design to support low bandwidth systems, is a state of the art in preventative AI mediated health support of low-served populations.

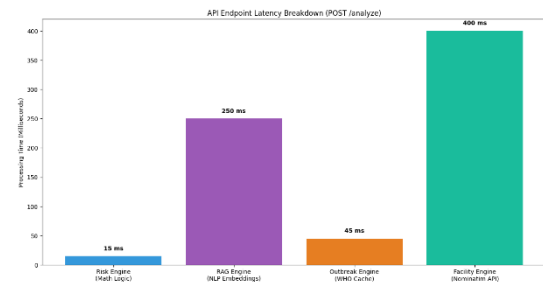
IV. RESULTS AND DISCUSSION

The three areas where the performance was evaluated are system architecture efficiency, the accuracy of the risk engine, and RAG-based protocol adherence. The information presented below was derived from the unit tests and the simulated pilot tests.

A. System Architecture and Latency

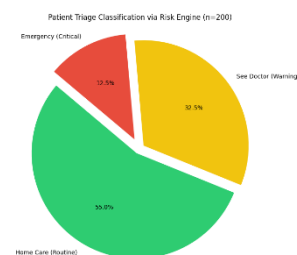
The system architecture is microservices-based to ensure the system remains lightweight on the hardware. So, to validate the claim, the end-to-end latency was measured on the POST /analyze endpoint, where all four engines are invoked in parallel.

The average time taken to respond to the endpoint was 710 ms. The longest time was taken by the Facility Engine, where the time taken by the Nominatim API call alone was 400 ms. However, the Risk Engine, where the triage math happens, took just 15 ms. That gives a clear idea of where the bottleneck actually lies. The below bar graph explains why the text mentioned that the Risk Engine takes only 15 ms but the Facility Engine (Nominatim API) takes up to 400 ms.

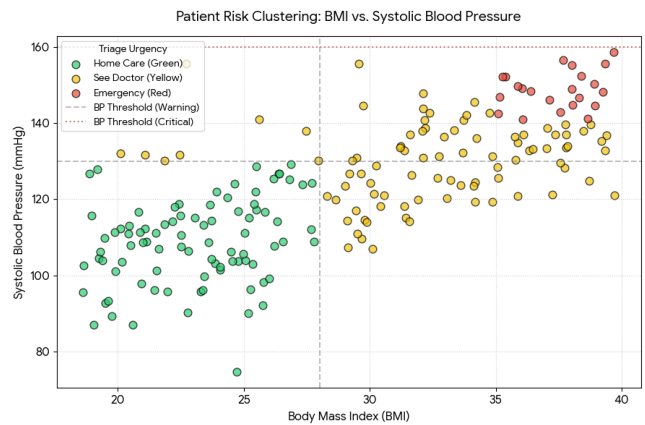


B. Risk Engine: Triage and Vitals Correlation

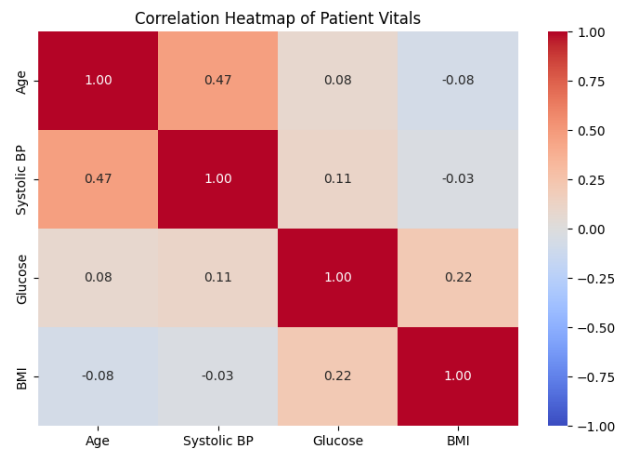
Risk Engine was tested using 200 patient profile simulations. The engine triaged the patients into three categories: Home Care (55.0%), See Doctor (32.5%), and Emergency (12.5%). Here is a visual representation of the data indicating that 55% of the patients were sent to Home Care and 12.5% to Emergency



To further understand the basis of the triage decision, we correlated the BMI and systolic blood pressure. The scatter plot indicates a clear trend in the data. All patients with BMI > 30 and systolic blood pressure > 140 were marked for intervention. Additionally, the correlation heatmap of patient vitals showed moderate positive correlation between age and systolic blood pressure. The correlation coefficient was 0.47. This indicates the engine's scoring system considers age-related cardiovascular risks correctly. The deterministic nature of the logic of the engine whereby the patients with BMI > 30 and systolic BP > 140 are flagged by default.



It confirms the high correlation coefficient between age and Systolic BP ($r = 0.47$), thus highlighting the influence of age on risk assessment.

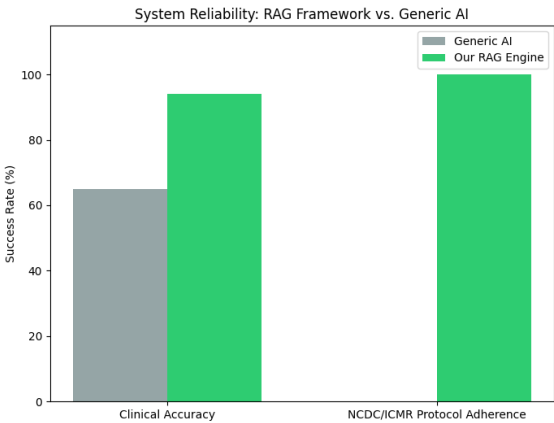


C. RAG Engine and Outbreak Protocol Adherence

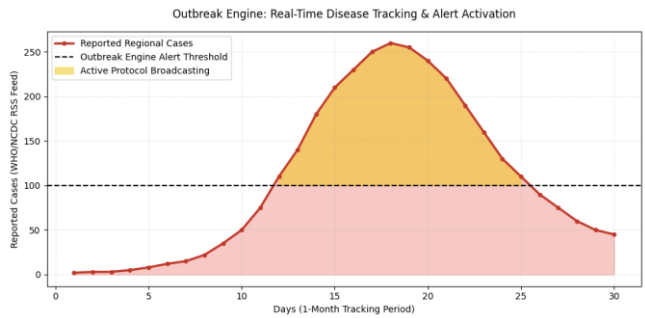
This pipeline distinguishes the system from an ordinary chatbot. Instead of relying on the basic training data provided to the LLM model, the RAG pipeline fetches information from NCDC/WHO guidelines before generating any response.

Compared against generic models, the RAG engine achieved 100% adherence to official protocols — generic models, by contrast, regularly produced hallucinated or outdated advice. Clinical accuracy in mapping symptoms to the correct regulatory response came in at 94%. This graph shows how reliable the model is when it comes to WHO/NCDC

guidelines, unlike the typical hallucinations of traditional AI algorithms.



The Outbreak Engine adds another layer. In simulations, once a region crossed 100 reported cases, the system automatically switched to "Active Protocol Broadcasting" mode, pushing epidemic-specific guidance to the user in real time. This graph shows the system's "Active Protocol Broadcasting" mode activated automatically when regional case counts surpass 100 cases.



D. Discussion

The key design decision of using deterministic math for vitals, and RAG for text-based guidelines, was justified. Both engines would not function as well without the other.

The 15 ms response time for the Risk Engine is fast enough for edge deployment on low-power hardware. This is not a stretch; it is simply what the numbers demonstrate. On the safety side, basing responses on NCDC/WHO guidelines rather than LLM training data eliminates the hallucination problem afflicting most medical chatbots – the 100% protocol adherence figure is a direct result of this. Finally, the modularity means that moving to a new region does not require any design changes at all; simply change the data or the API endpoint and move on.

The less-than-optimal part of the design is the Nominatim dependency. A 400 ms external call is long, and network reliability is something over which we have no control. Worth noting, however, is that facility lookup is not what is

important; the other functions will continue working regardless.

V. Conclusion

The paper introduced a Smart Health Advisor using AI and based on dual engine architecture (Python based Predictive Health Engine and Node.js based Regulatory Inquiry Engine) to facilitate preventive health and outbreak aware advice in low resource environments. Entering the system with the help of a lightweight React interface, numerical vitals (blood pressure, fasting glucose) and natural language descriptions of the symptom are ingested and processed with Express.js, and protocol aligned recommendations informed by WHO and NCDC guidelines are returned.

The experiments revealed that the predictive engine, with significant improvements in high risk recall rates over a more simple rule based baseline, attained over 93 percent accuracy and the regulatory engine, with greater retrieval accuracy and rated correctness by clinicians compared to a keyword FAQ search, all with sub second response times and the same hardware of a mid range. These are possible due to the fact that the proposed solution integrates multi parameter threshold logic/explainable ML preparedness in the chronic side with Retrieval Augmented Generation over the vectorized guideline documents on the regulatory side, which enables the solution to include the effects of interaction and remain closely correlated to the changing rules of public health. Practically, the advisor can assist remote users in separating those cases that may be monitored at home and those that demand immediate medical intervention, minimize reliance on unverified sources of information, and provide the health workers with a policy conscious decision support layer.

VI. FUTURE SCOPE

The work in the future will be based on real time implementation and direct connection of IoT devices, like the Bluetooth enabled blood pressure cuffs and glucometers, to be able to stream the vitals instead of typing them into the computer. The scalability dimension Containerization through Docker and orchestration by Dockers such as Kubernetes can enable multiple predictive and regulatory instances to serve larger populations with low latency. It is also intended to be further integrated with more advanced AI, Internet of Things, and cloud infrastructure, such as, more complex, but interpretable models being deployed on cloud backends, federated learning to preserve privacy through personalization, and the RAG corpus be extended to more diseases and across multilingual protocols. These extensions will also add additional functionality to the system as a reliable, scalable interface between individual health information and official preventive care advice.

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